JONGWOOK JEON

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Education

Yonsei University Mar 2020 – Present
School of Electrical and Electronic Engineering
GPA: 3.83/4.5
Leave of absence for military service: May 2022 - Nov 2023

Research Interests

My research interests center on building generalizable generative policies that can adapt across offline datasets, online interaction, and long-horizon real-world tasks. In particular, I focus on two directions:

- **RL Fine-tuning for Generative Policy Models:**
  improving imitation-learned policies through RL fine-tuning, with a focus on flow-based and diffusion-based policy models.
- **Skill-based and Hierarchical Reinforcement Learning:**
  composing reusable skills for long-horizon tasks, especially in VLA models and flow-/diffusion-based policies.

Publications

* indicates equal contribution.

Learning Multimodal One-step Flow Policy via Value-weighted Optimal Transport

Jaehun Shon*, Jinha Choi*, **Jongwook Jeon**, Jongmin Lee (Under Review)

- Proposed a value-weighted optimal transport framework for learning efficient one-step flow policies in offline reinforcement learning.
- Contributed to algorithmic discussions on model design, performance improvement, and conducted baseline experiments.

MIMDICE: Unsupervised Skill Discovery from Arbitrary Off-policy Experience

Younghyun Kim, **Jongwook Jeon**, Jongmin Lee (Under Review)

- Proposed a DICE-based approach to reduce replay-buffer bias in mutual-information-based skill discovery.
- Contributed to baseline experiments and surveyed the preliminaries and derivation verification.

Experience

Yonsei DILLAB , Reserach Intern (Adivsor: prof. Jongmin Lee) Apr 2025 – Present

- Co-authored a study on value-weighted optimal transport for offline flow policy learning.
- Co-authored a study on DICE-based skill discovery for mutual-information-based skill learning.

Yonsei CVLAB , Reserach Intern (Adivsor: prof. Bumsub Ham) Jan 2025 – Apr 2025

- Presented and discussed weekly on essential papers in AI and computer vision

Yonsei Data Science Lab , Vice President (Adivsor: prof. Taeyoung Park) Jul 2024 – Jun 2025

- Studied fundamental machine learning theory (CV, NLP, Diffusion etc)
- Conducted modeling projects (Optimizing Learning Rate Scheduler via RL, Speech splitting to Native Accent) in my areas of interest, including industry-sponsored projects (Pet's body length Measurement via 3d vision)

Projects

Text-Aware Image Restoration via Reinforcement Learning Mar 2026 - Current

- Proposed and currently investigating an online RL fine-tuning framework for text-aware image restoration, motivated by the misalignment between restoration metrics and human preference in text rendering.

AI Pharmacy Copilot for Robotic Medication Sorting, Physical AI Hackathon — Mar 2026

- Built a robotic assistant for medication sorting using Korean STT, RGB-D perception, VLM-based reasoning, and ROS2-based execution.
- Contributed to testing VLMs such as *Cosmos-Reason2* and *Qwen3*, integrated the STT module, and explored Isaac Sim-based RL training for improved robustness and autonomy.

Optimizing Optimizer and Learning rate scheduler, team leader — Mar 2025 – May 2025

- Formulated optimizer and learning rate scheduler design as a sequential decision-making problem and explored reinforcement-learning-based update rules.
- Contributed to Ideation and optimizing via DQN approach.

Finance Chatbot, team leader — Sep 2024 – Dec 2024

- Built a GenAI-powered finance chatbot to improve financial literacy among college students, using a RAG pipeline with hourly news database updates.
- Contributed to task-specific development using the OpenAI API, focusing on backend engineering (DB, data crawling, and RAG)

Awards

AI x Robotics Hackathon (Grand Prize) — Mar 2026

Awarded the Grand Prize at the AI x Robotics Hackathon organized by XYZ Company for developing an AI-powered robotic assistant for medication sorting and organization.

- Contributed to the system design and integration of speech, vision-language reasoning, and robotic control.

Yonsei 2nd Gen AI (Silver Prize) — Sep 2024 - Dec 2024

Awarded Silzer Prize in the Gen AI contest making Finance Chatbot web service.

- Positioned as backend engineer and adjusting Open AI API.

Relavant Courseworks

AI and Machine Learning: Introduction to Artificial, Intelligence, Reinforcement Learning, Generative Model, Robot Learning, Intellgent Control

Mathemateics: Probability and Statistics, Number Theory, Engineering Mathmatics, Linear Algebra, Optimization, Analysis1

Systems and Computing: Digital Signal Proccessing, Data structure, Computer Architecture, Operating Systems

Skills

Programming Languages: Python/Pytorch, C/C++, Linux Environment, MATLAB

Languages: Korean(Native), English(Proficient)

Activities: Yonsei Autonomous eXtenstion (YAX), Yonsei Futsal and Soccer club (YFS)